



Project Proposal – R.E.P. Tracker

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The Problem:

Weightlifting is a very daunting task and can be dangerous if done incorrectly. For many people, the gym is sometimes too complex and confusing. For others, there is too much research that must be done to find the appropriate weight to achieve maximum gain. The R.E.P. (**R**epetition. **E**fficiency. **P**recision.) Tracker system helps to resolve these issues by tracking exercises, monitoring lifting form, and recording training metrics for long-term analysis. Furthermore, the R.E.P. Tracker is customizable to the user and compiles data and software to calculate various performance-based metrics in contribution to tracking and trending.

Proposed Solution:

First, we address incorrect form with our Motional Tracking in conjunction with the Performance Monitoring System. Respectively, these subsystems will record the user's reps, sets, and angles performed while measuring overall work output and exercise intensity (via the user's oxygen levels, temperature, heart rate, etc...). With this data, they will coordinate with User Feedback to inform the athlete on what is being done incorrectly while utilizing real-time performance tracking. These subsystems will notify you of bad form or other inefficiencies in the workout to avoid acute injury and/or development of potentially hindering habits. Simultaneously, the Adjustable Camera Feed will help in a similar way by showing the user what the problem is visually and allowing them to adjust as needed without the use of a coach or gym partner. The complexity and confusion of a proper "rep" is removed as this adjustment process records the exercise metrics (Motional Tracking, User Interface, Performance Monitoring) and suggests corrections to improper form while allowing them to adjust until satisfactory performance is achieved in real-time (Camera-Feedback). The Programmable Graphical User Interface functions via processing user recorded basic information (e.g. bodyweight, height, fitness level, etc...). This basic information sets the foundation for the range of exercise specific metrics that the User Feedback subsystem utilizes. For example, a hypothetical user of low fitness level tries to squat 500lbs... the User Feedback system will notify them of their potentially devastating mistake. Finally, the lack of data and research availability for the user will be solved with all previous subsystems providing statistics for tracking and trending (graphs, charts, etc...). These metrics allow the R.E.P. Tracker to provide the exerciser with comprehensible form analysis (trending negatively or positively), long-term progress (daily/weekly/monthly), and suggestions for planning workouts.

System Diagram:

